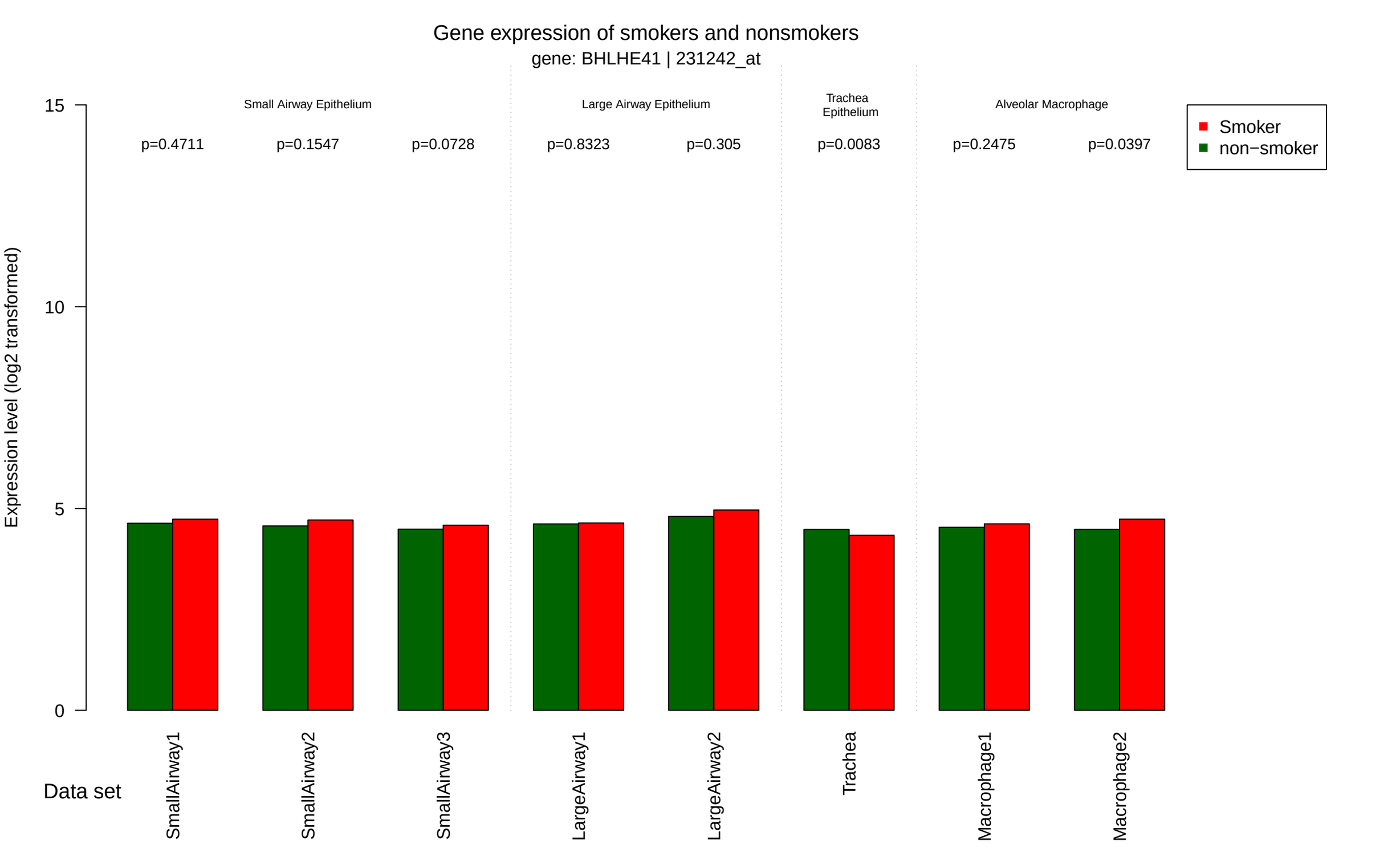
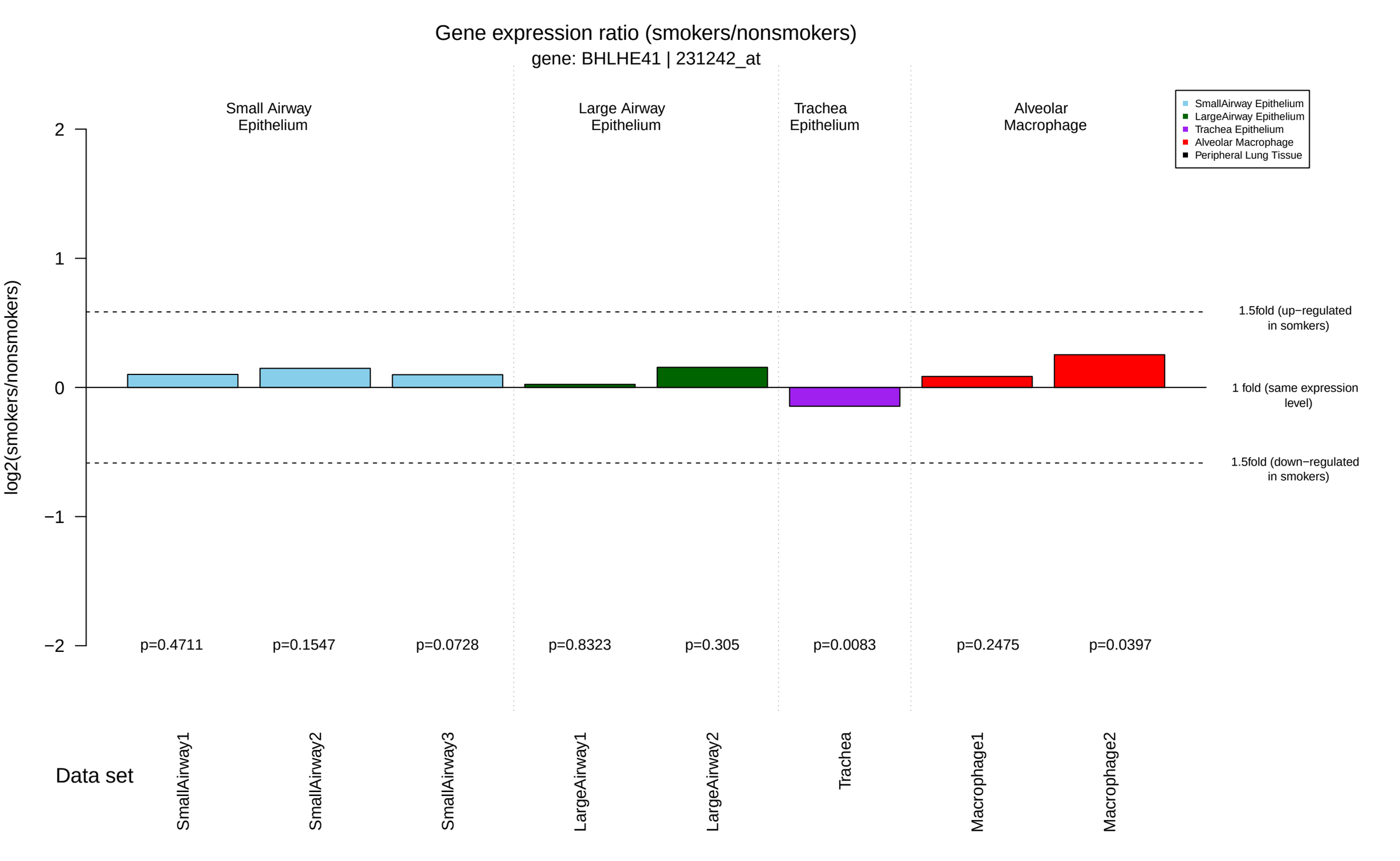
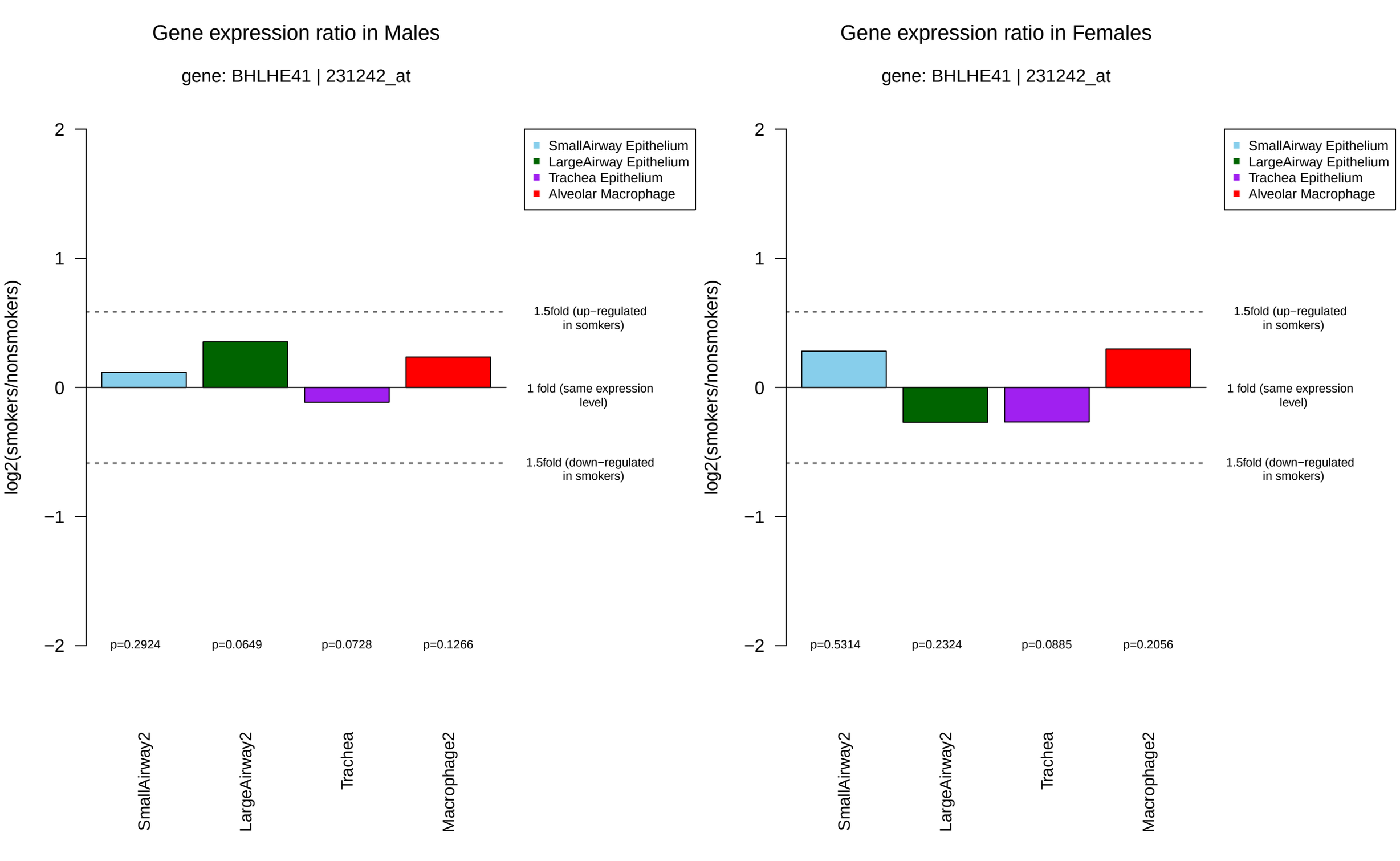
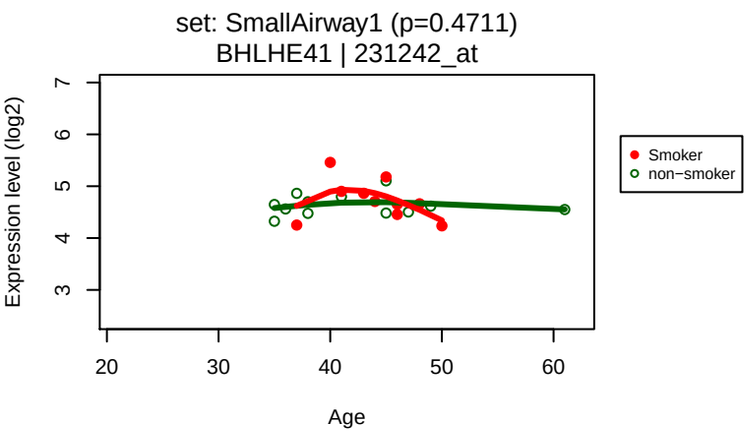
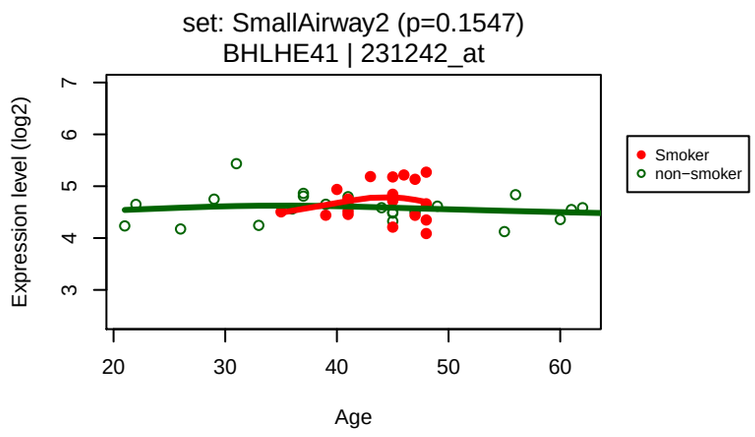
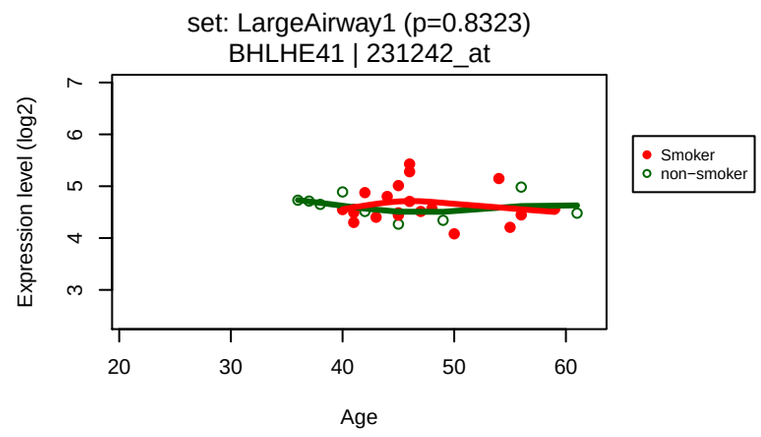

Gene expression of smokers and nonsmokers
gene: BHLHE41 | 231242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4711
p=0.1547
p=0.0728
p=0.8323
p=0.305
p=0.0083
p=0.2475
p=0.0397
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

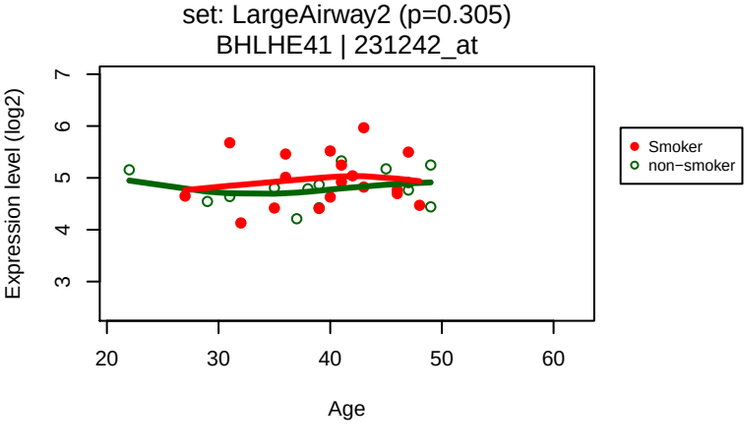

Gene expression ratio (smokers/nonsmokers)
gene: BHLHE41 | 231242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4711
p=0.1547
p=0.0728
p=0.8323
p=0.305
p=0.0083
p=0.2475
p=0.0397
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

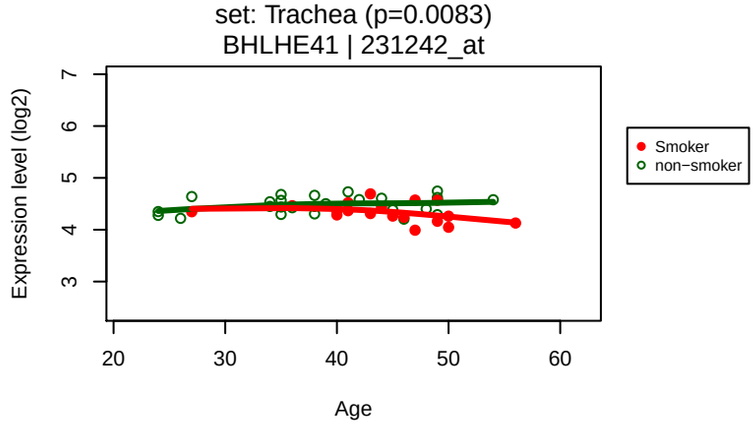

Gene expression ratio in Males
gene: BHLHE41 | 231242_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2924
p=0.0649
p=0.0728
p=0.1266
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BHLHE41 | 231242_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5314
p=0.2324
p=0.0885
p=0.2056
SmallAirway2
LargeAirway2
Trachea
Macrophage2

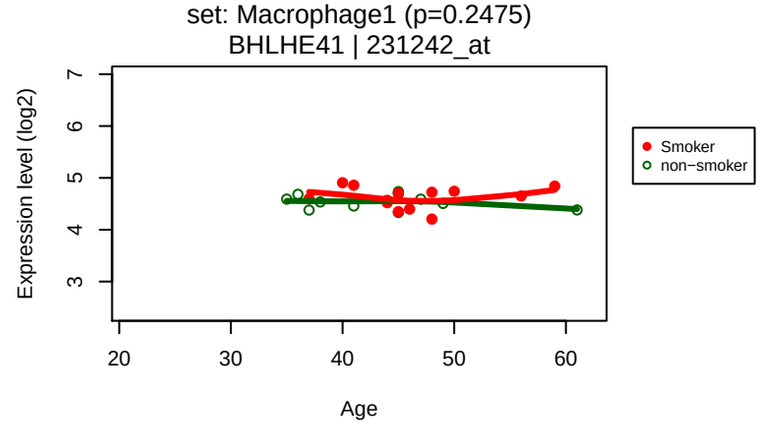

set: SmallAirway1 (p=0.4711)
BHLHE41 | 231242_at
Expression level (log2)
Age
Smoker
non−smoker

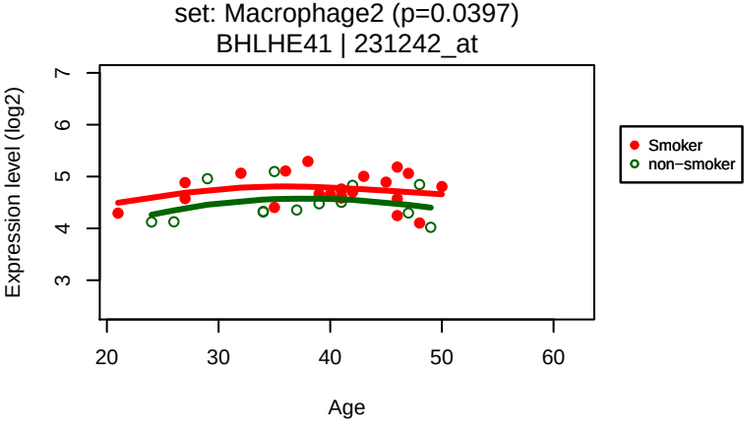

set: SmallAirway2 (p=0.1547)
BHLHE41 | 231242_at
Expression level (log2)
Age
Smoker
non−smoker

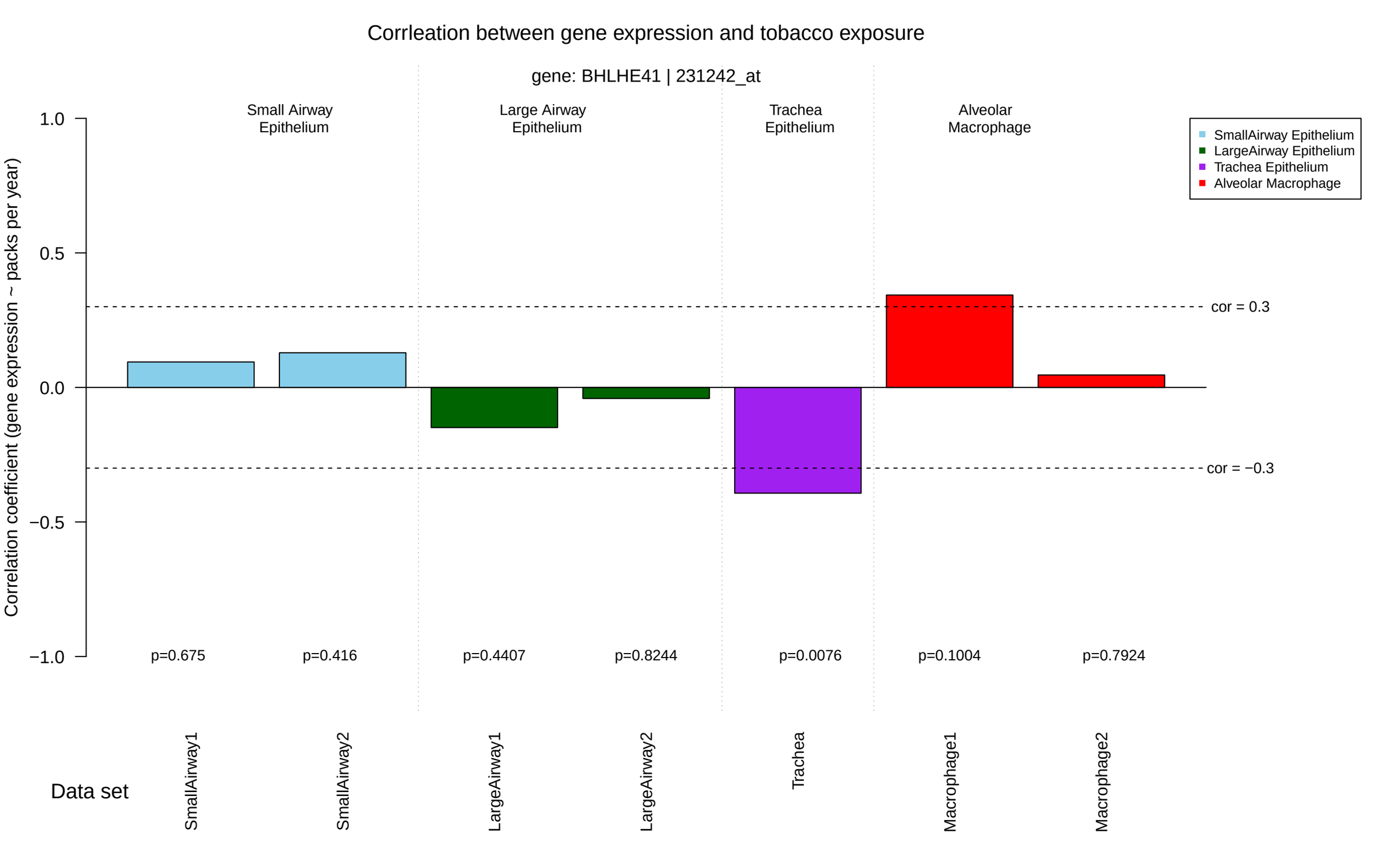

set: LargeAirway1 (p=0.8323)
BHLHE41 | 231242_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.305)
BHLHE41 | 231242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0083)
BHLHE41 | 231242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2475)
BHLHE41 | 231242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0397)
BHLHE41 | 231242_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BHLHE41 | 231242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.675
p=0.416
p=0.4407
p=0.8244
p=0.0076
p=0.1004
p=0.7924
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2